moonsize

WEBASSEMBLY / SIZE ANALYSIS

# WebAssembly size report

`D:\project\moonsize\examples\fib.wasm`

47 local functions · 10 module labels · 81 modeled references

**File size**

## 10,675 B

10.4 KB · complete artifact

**Code section**

## 5,166 B

48.3% of file · complete encoding

**Whole-file gzip**

## 5,207 B

48.7% of raw · 2.05x

**Unreachable in model**

## 0 B

no unreachable local function bytes

**No detected graph gaps**

No gaps detected within the current reference model. This is not full Wasm verification.

**Model results identify investigation candidates, not guaranteed deletion savings.**

Unreachable functions are not a safe-deletion list. Read the model and measurement limits.

Chart library and report data are embedded in this file; no network or sibling assets are required.

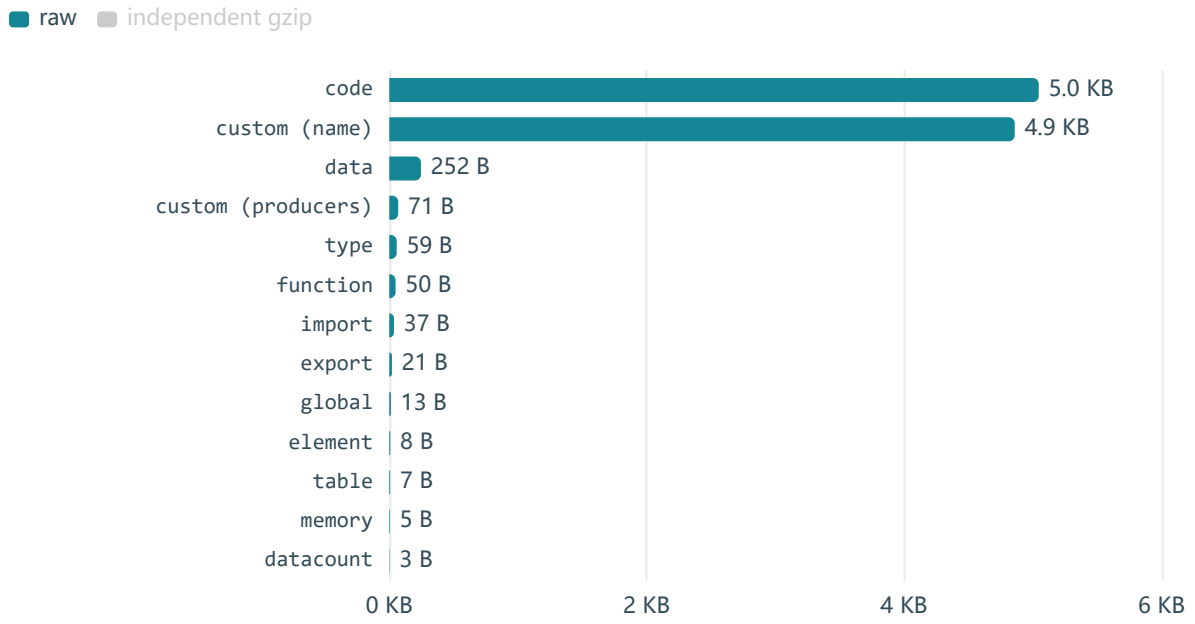
# Where the bytes go

Complete sections, then attributed function encodings

## Sections

Raw sizes include section headers and payloads. Both views use the same byte axis.

Each section is compressed separately. These sizes are not additive and do not represent whole-file transfer shares.




## Modules • attributed function bytes

Shares are based on attributed function bytes; code-section overhead is excluded. Small modules may be grouped as other.



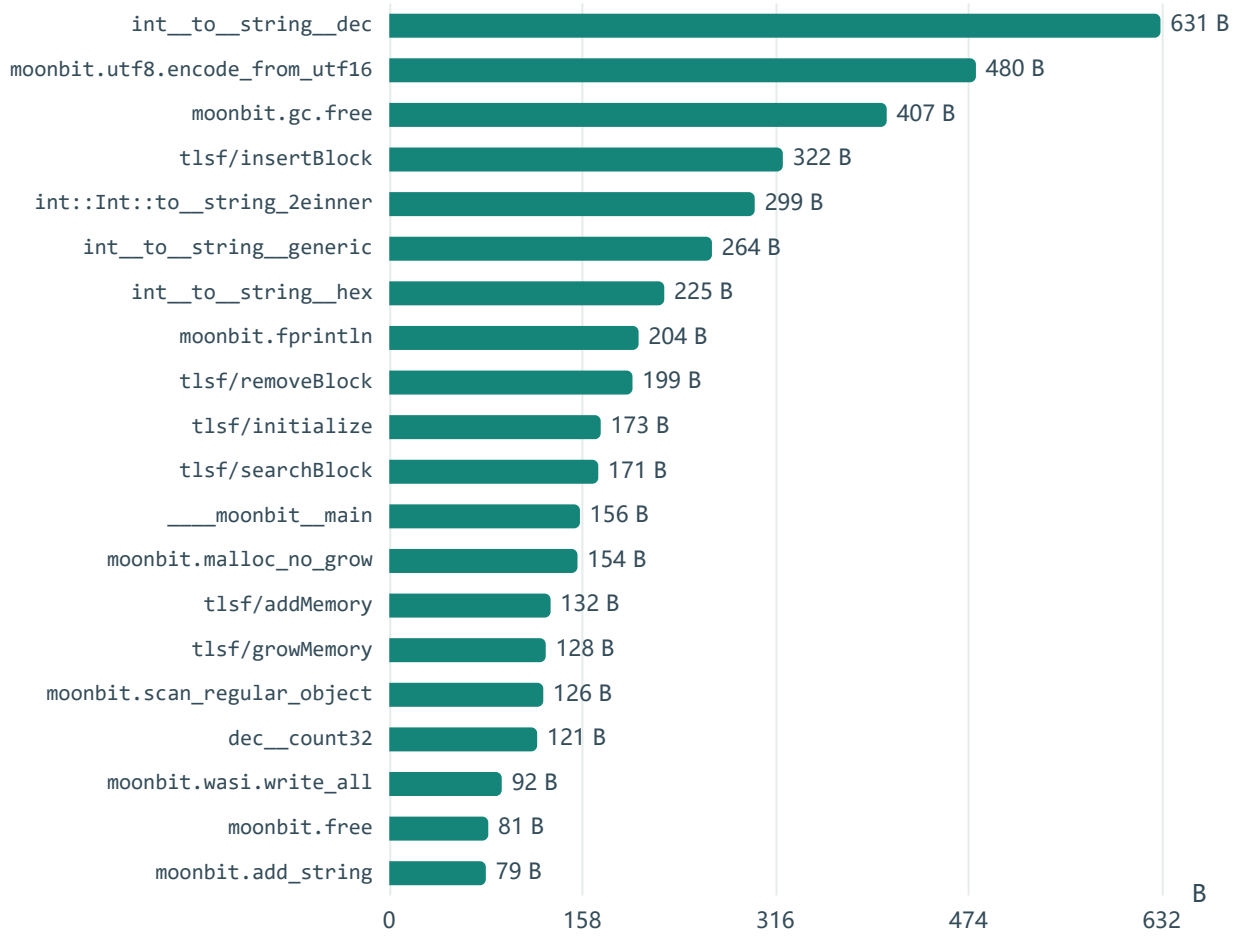
(top-level) tlsf moonbit moonbit.utf8 moonbit.gc int::Int moonbit.wasi BigSaltyM 1/2

(top-level) · 1,535 B
moonbit · 1,030 B
moonbit.gc · 439 B
moonbit.wasi · 111 B
other (2) · 13 B
tlsf · 1,175 B
moonbit.utf8 · 482 B
int::Int · 301 B
BigSaltyMan::moonsize::examples · 76 B

## Functions to investigate

Top 20 local functions, ranked by body bytes

Function gzip uses independent streams including size prefixes, with ratios based on encoded size. It does not measure transfer savings from removal.




### ▼ Function details • full names and measured bytes

Same Top 20 local functions as the body-size chart. Encoded B includes the size prefix. Indices use the full function index space, including imports. Reachability is within the current model.

| Rank | Index | Function / module | Body B | Encoded B | Independent gzip B | Model reachability |
|---|---|---|---|---|---|---|
| 1 | 43 | **int__to__string__dec**<br>(top-level) | 631 | 633 | 385 | Reachable in model |
| 2 | 21 | **moonbit.utf8.encode_from_utf16**<br>moonbit.utf8 | 480 | 482 | 268 | Reachable in model |
| 3 | 29 | **moonbit.gc.free**<br>moonbit.gc | 407 | 409 | 254 | Reachable in model |
| 4 | 3 | **tlsf/insertBlock**<br>tlsf | 322 | 324 | 220 | Reachable in model |

| Rank | Index | Function / module | Body B | Encoded B | Independent gzip B | Model reachability |
|---|---|---|---|---|---|---|
| 5 | 39 | **int::Int::to__string_2einner**<br>int::Int | 299 | 301 | 218 | Reachable in model |
| 6 | 44 | **int__to__string__generic**<br>(top-level) | 264 | 266 | 211 | Reachable in model |
| 7 | 45 | **int__to__string__hex**<br>(top-level) | 225 | 227 | 180 | Reachable in model |
| 8 | 33 | **moonbit.fprintln**<br>moonbit | 204 | 206 | 183 | Reachable in model |
| 9 | 2 | **tlsf/removeBlock**<br>tlsf | 199 | 201 | 174 | Reachable in model |
| 10 | 5 | **tlsf/initialize**<br>tlsf | 173 | 175 | 160 | Reachable in model |
| 11 | 1 | **tlsf/searchBlock**<br>tlsf | 171 | 173 | 144 | Reachable in model |
| 12 | 47 | **____moonbit__main**<br>(top-level) | 156 | 158 | 147 | Reachable in model |
| 13 | 8 | **moonbit.malloc_no_grow**<br>moonbit | 154 | 156 | 136 | Reachable in model |
| 14 | 4 | **tlsf/addMemory**<br>tlsf | 132 | 134 | 136 | Reachable in model |
| 15 | 7 | **tlsf/growMemory**<br>tlsf | 128 | 130 | 139 | Reachable in model |
| 16 | 30 | **moonbit.scan_regular_object**<br>moonbit | 126 | 127 | 124 | Reachable in model |
| 17 | 42 | **dec__count32**<br>(top-level) | 121 | 122 | 110 | Reachable in model |
| 18 | 32 | **moonbit.wasi.write_all**<br>moonbit.wasi | 92 | 93 | 101 | Reachable in model |
| 19 | 10 | **moonbit.free**<br>moonbit | 81 | 82 | 97 | Reachable in model |
| 20 | 26 | **moonbit.add_string**<br>moonbit | 79 | 80 | 94 | Reachable in model |

## Module map

Module → function · raw encoded bytes

Area is raw encoded bytes. Tooltip gzip values are independent function streams, not additive transfer shares. Modules keep their ring colors; outlined cells and text identify functions unreachable in the model.




### ▼ Methods, measurement ranges & model limits

#### Reference model

Direct calls, including direct tail calls, use exact targets; table calls use static element candidates. Reference-call targets and dynamic table contents are not solved. Imported/exported tables, runtime writes and explicit initializers are conservatively diagnosed, even with static candidates. Element references are conservative roots, including private, passive and declarative segments. graph_complete means no gaps detected within this scope, not full Wasm certification. RETAINED sums dominated encoded function bytes within the current graph/root model. It does not predict recompiled file reduction or whole-file gzip savings, even when --strict passes.

RETAINED sums dominated encoded function bytes within the current graph/root model. It does not predict recompiled file reduction or whole-file gzip savings, even when --strict passes.

#### Compression

Whole-file gzip is one complete-file stream. Sections include their headers and payloads; functions include their size prefixes (size, not bodySize). Each local range is an independent gzip stream. These sizes are not additive and do not represent whole-file transfer shares or removal savings.

#### Chart availability

ECharts, styles, scripts and measured data are embedded in this file. No remote fonts, chart assets or sibling files are required; the report can be moved to another directory.

Generated by moonsize · 47 local functions. Model findings guide investigation; verify optimization outcomes by rebuilding and comparing complete artifacts.